

NORAIZ RANA

Task #14

Topic

Styling in React and
Component Architecture

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APPVERSE TECHNOLOGIES

Learning Report: Styling in React and Component Architecture

1. Introduction

Styling is a vital part of frontend development, especially when working with component-based libraries like **React**. As applications grow, maintaining clean, scalable, and modular styles becomes essential.

In this learning module, I explored **three modern approaches to styling React applications**:

- **CSS Modules**
- **Tailwind CSS**
- **Styled Components**

Each approach offers unique advantages depending on the project requirements, scale, and development preferences.

2. Component Architecture in React

React promotes **modular and reusable architecture**, where the UI is built using independent components. Each component ideally contains:

- Its **own logic**
- Its **own styling**
- Isolated **data and behavior**

Organizing components this way improves **reusability, scalability, and maintainability**.

3. CSS Modules

CSS Modules allow you to write **locally scoped CSS**. Unlike global CSS, class names are automatically scoped to the component.

Key Benefits:

- No class name conflicts
- Familiar CSS syntax
- Works with existing tooling

Example:

```
import styles from './Button.module.css';

function Button() {
  return <button
    className={styles.primary}>Click Me</button>;
}
/* Button.module.css */
.primary {
  background-color: #3498db;
  padding: 10px 20px;
}
```

4. Tailwind CSS

Tailwind CSS is a **utility-first CSS framework** that allows rapid styling by applying utility classes directly to HTML/JSX elements.

Key Benefits:

- Highly customizable via configuration
- Encourages consistency and design systems
- No need to leave JSX to style

Example:

```
function Card() {
  return (
```

```
    <div className="bg-white shadow-md p-4
rounded-lg">
      <h2 className="text-lg font-bold">Tailwind
Card</h2>
    </div>
  );
}
```

Tailwind reduces the need for custom CSS files and speeds up development.

5. Styled Components

Styled Components is a popular library that allows writing **CSS-in-JS**, where styles are defined inside the component file using template literals.

Key Benefits:

- Dynamically style components using props
- Encapsulated styling per component
- JavaScript logic + styles in one file

Example:

```
import styled from 'styled-components';

const Button = styled.button`
  background: #2ecc71;
  color: white;
  padding: 10px 20px;
`;

function SubmitButton() {
  return <Button>Submit</Button>;
}
```
